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Java SE 27 & JDK 27

Class MethodHandleProxies

java.lang.Object
    java.lang.invoke.MethodHandleProxies

```
public final class MethodHandleProxies
extends Object
```

This class consists exclusively of static methods that help adapt method handles to other JVM types, such as interfaces.

Since:

1.7

Method Summary

All Methods	Static Methods	

Modifier and Type	Method	Description
static <T> T	**asInterfaceInstance**(Class<T> intfc, MethodHandle target)	Produces an instance of the given single-method interface which redirects its calls to the given method handle.
static boolean	**isWrapperInstance**(Object x)	Determines if the given object was produced by a call to asInterfaceInstance.
static MethodHandle	**wrapperInstanceTarget**(Object x)	Produces or recovers a target method handle which is behaviorally equivalent to the unique method of this wrapper instance.
static Class<?>	**wrapperInstanceType**(Object x)	Recovers the unique single-method interface type for which this wrapper instance was created.

Methods declared in class Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Method Details

asInterfaceInstance

```
public static <T> T asInterfaceInstance(Class<T> intfc,
                                        MethodHandle target)
```

Produces an instance of the given single-method interface which redirects its calls to the given method handle.

A single-method interface is an interface which declares a uniquely named method. When determining the uniquely named method of a single-method interface, the public `Object` methods (`toString`, `equals`, `hashCode`) are disregarded as are any default (non-abstract) methods. For example, `Comparator` is a single-method interface, even though it re-declares the `Object.equals` method and also declares default methods, such as `Comparator.reverse`.

The interface must be public, not hidden, and not sealed. No additional access checks are performed.

The resulting instance of the required type will respond to invocation of the type's uniquely named method by calling the given target on the incoming arguments, and returning or throwing whatever the target returns or throws. The invocation will be as if by `target.invoke`. The target's type will be checked before the instance is created, as if by a call to `asType`, which may result in a `WrongMethodTypeException`.

The uniquely named method is allowed to be multiply declared, with distinct type descriptors. (E.g., it can be overloaded, or can possess bridge methods.) All such declarations are connected directly to the target method handle. Argument and return types are adjusted by `asType` for each individual declaration.

The wrapper instance will implement the requested interface and its super-types, but no other single-method interfaces. This means that the instance will not unexpectedly pass an `instanceof` test for any unrequested type.

Implementation Note: Therefore, each instance must implement a unique single-method interface. Implementations may not bundle together multiple single-method interfaces onto single implementation classes in the style of `AWTEventMulticaster`.

The method handle may throw an *undeclared exception*, which means any checked exception (or other checked throwable) not declared by the requested type's single abstract method. If this happens, the throwable will be wrapped in an instance of `UndeclaredThrowableException` and thrown in that wrapped form.

Like `Integer.valueOf`, `asInterfaceInstance` is a factory method whose results are defined by their behavior. It is not guaranteed to return a new instance for every call.

Because of the possibility of bridge methods and other corner cases, the interface may also have several abstract methods with the same name but having distinct descriptors (types of returns and parameters). In this case, all the methods are bound in common to the one given target. The type check and effective `asType` conversion is applied to each method type descriptor, and all abstract methods are bound to the target in common. Beyond this type check, no further checks are made to determine that the abstract methods are related in any way.

Future versions of this API may accept additional types, such as abstract classes with single abstract methods. Future versions of this API may also equip wrapper instances with one or more additional public "marker" interfaces.

Type Parameters:

`T` - the desired type of the wrapper, a single-method interface

Parameters:

`intfc` - a class object representing `T`

`target` - the method handle to invoke from the wrapper

Returns:

a correctly-typed wrapper for the given target

Throws:

`NullPointerException` - if either argument is null

`IllegalArgumentException` - if the `intfc` is not a valid argument to this method

`WrongMethodTypeException` - if the target cannot be converted to the type required by the requested interface

isWrapperInstance

```
public static boolean isWrapperInstance(Object x)
```

Determines if the given object was produced by a call to `asInterfaceInstance`.

Parameters:

x - any reference

Returns:

true if the reference is not null and points to an object produced by `asInterfaceInstance`

wrapperInstanceTarget

```
public static MethodHandle wrapperInstanceTarget(Object x)
```

Produces or recovers a target method handle which is behaviorally equivalent to the unique method of this wrapper instance. The object x must have been produced by a call to `asInterfaceInstance`. This requirement may be tested via `isWrapperInstance`.

Parameters:

x - any reference

Returns:

a method handle implementing the unique method

Throws:

`IllegalArgumentException` - if the reference x is not to a wrapper instance

wrapperInstanceType

```
public static Class<?> wrapperInstanceType(Object x)
```

Recovers the unique single-method interface type for which this wrapper instance was created. The object x must have been produced by a call to `asInterfaceInstance`. This requirement may be tested via `isWrapperInstance`.

Parameters:

x - any reference

Returns:

the single-method interface type for which the wrapper was created

Throws:

`IllegalArgumentException` - if the reference x is not to a wrapper instance

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.

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